

LEVERAGING MULTILAYER PERCEPTRONS TO COMPUTE GRADIENTS FOR IDENTIFYING GENES ASSOCIATED WITH BREAST CANCER

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Abstract

Breast cancer is a continuing global issue, with new patients being diagnosed with the disease daily. It remains a serious concern due to its high mortality rate and widespread impact. With the rise of new technologies, this study aims to leverage artificial intelligence in the medical field to identify the genes most commonly associated with breast cancer. By identifying these genes, future researchers can focus on key genetic markers to improve prevention and treatment.

This study uses the GTEx and TCGA datasets, widely used and trusted sources in genomic research, to analyze gene expression across tissue samples. After processing the genomics dataset using a log2 transformation, the training data is fed into a multilayer perceptron (MLP) classification model. A gradient-based attribution method is then used to compute gradients with respect to the 19,738 gene features using backpropagation. Gradients are computed to rank genes that have the greatest influence on the model's predictions by examining how changes in each input feature affect the MLP's output. A baseline MLP model is first applied using the full set of gene expression features. Afterward, a silhouette score-based feature selection approach is used to identify more important features for a more focused pre-processing step.

After identifying and ranking the genes with the greatest influence on the prediction, the results are further analyzed. First, a Fisher's exact test is performed on a random gene set to ensure the results are not due to random chance. Next, the same test is conducted using a gene set already associated with breast cancer to check for a meaningful correlation within the identified gene list. Once the gene list shows a clear signal, it is further analyzed using gene enrichment to find significant biological categories related to breast cancer. The results identify a gene set associated with breast cancer, allowing scientists to focus on specific genes while also demonstrating the usefulness of artificial intelligence in biomedical research.

Table of Contents

Title Page	i
Abstract	ii
List of Tables	iv
List of Figures	v
1 Introduction	1
2 Research Design and Methods	3
2.1 Data Retrieval and Preprocessing	3
2.2 Multilayer Perceptron (MLP) Classification Model	4
2.3 Gradient-Based Feature Attribution Model	6
2.4 Feature Selection	8
2.5 Evaluation Methods	10
3 Related Work	14
3.1 Gene signatures provides prognostic information in breast cancer	14
3.2 Biomarker Genes with enriched classification from Hallmark gene sets	15
3.3 Gene Selection and Classification of Cancer using Deep Learning	15
4 Results	16
4.1 MLP Classification Model Accuracy	16
4.2 Utilizing all 19,738 Gene Features - MLP307	18
4.3 Utilizing Gene Features from C4, C6, and Hallmark Gene Sets	23
5 Conclusions and Discussion	28
5.1 Interpretation of Evaluation Findings	28
5.2 Limitations	29
5.3 Future Work	30
5.4 Concluding Remarks	30
5.5 Acknowledgments	30
Bibliography	31

List of Tables

2.1	Contingency table used for Fisher's exact test.	11
4.1	Test set performance metrics for the MLP classifier using 19,738 features	16
4.2	Test set performance metrics for the MLP classifier using 4,311 features extracted from the C4 gene set.	17
4.3	Test set performance metrics for the MLP classifier using 3,311 features extracted from the C6 gene set.	17
4.4	Test set performance metrics for the MLP classifier using 968 features extracted from the Hallmark gene set.	17
4.5	Fisher's exact test results comparing overlap between MLP307, Random307, and Core307.	20
4.6	Fisher's exact test results comparing random gene sets (Random307) with Core307 and other random sets.	20
4.7	Fisher's exact test results comparing threshold-based gene sets with Core307 and Random307.	22
4.8	Fisher's exact test results comparing overlap between gene sets C4, C6, and Hallmark against Random307 and Core307.	24
4.9	Top enriched disease terms in C4 gene set.	25
4.10	Top enriched disease terms in C6 gene set.	25
4.11	Top enriched disease terms in Hallmark gene set.	25
4.12	Protein-Protein Interaction between C4 and C6	26

List of Figures

2.1	Example of MLP network "feed-forward" [12]	4
2.2	Distribution of Gene Importance Scores	7
4.1	Training and Validation Loss vs. Epoch (C4 gene set)	18
4.2	Venn Diagram comparing the gene sets of MLP307, Core307, and Random307	19
4.3	Comparison of Gene Sets Selected from 19,738 Genes Above a Threshold Against Core307 and Random307	21
4.5	Venn Diagram comparing the gene sets of C4, C6, and Hallmark	24
4.6	Connectivity distributions compared to random gene sets	27

Chapter 1

Introduction

Despite the overall decrease of the breast cancer mortality rate, approximately 1 in 8 women in the United States, or 13%, will be diagnosed with breast cancer. In addition, around 1 in 43 of women, or 2%, will die from the disease. Not only is the risk of death still high, but the added cost of chemotherapy, surgery, and the people it impacts are huge factors when assessing the burden of the disease [6]. Breast cancer is a heterogeneous malignancy, complex and highly variable in how it affects different patients. In addition, this disease can also vary within a single tumor mass. The disease typically originates in epithelial cells and is driven by genetic and epigenetic mutations in normal breast tissue [17].

Due to the complex and unique nature of breast cancer, there is a strong need for computational methods in the biomedical field to identify harmful genetic mutations. However, gene expression datasets are extremely high-dimensional, often containing tens of thousands of gene features, making it difficult to identify which genes are truly relevant. Traditional methods may struggle with noise, redundancy, and the scale of the data.

This thesis aims to utilize Artificial Intelligence, specifically a multilayer perceptron (MLP) classification model, on genomics datasets to identify genes that strongly correlate with breast cancer. By leveraging high-performance computing (HPC), the model can process large-scale datasets containing up to 19,738 gene features and efficiently determine the importance of individual genes. Neural networks allow for rapid identification of key features, reducing the time and effort required compared to traditional approaches.

To ensure that the selected genes are biologically meaningful, this study incorporates mul-

multiple validation methods, including Venn diagrams, Fisher’s exact test, enrichment analysis, and protein–protein interaction analysis. These methods are used to evaluate whether the identified gene sets align with known biological pathways and established breast cancer gene sets.

The contributions of this thesis include a computational pipeline using an MLP classification model to identify important gene features from a high-dimensional genomic dataset. Second, the pipeline integrates feature selection with statistical validation methods to evaluate whether the identified genes are biologically meaningful. The analysis includes creating Venn diagrams and applying Fisher’s exact test to determine whether there is significant overlap with a pre-established gene set and minimal overlap with a random gene set. It then performs enrichment analysis and protein–protein interaction analysis to evaluate whether the identified gene set shows biological relevance to breast cancer and proper connectivity within the set. Overall, this work establishes a foundation for combining AI-based approaches with biological analysis to study disease and gene expression.

Chapter 2

Research Design and Methods

This section describes the workflow’s overall design and implementation. The program operates in the following stages: (1) retrieval and preprocessing of genomics data, (2) application of a multilayer perceptron (MLP) classification model, (3) generation of gradient-based feature attribution, (4) feature selection and extraction, and (5) evaluation methods.

2.1 Data Retrieval and Preprocessing

The goal of this research is to identify specific genes that the MLP considers significant for breast cancer classification. Before the data can be fed into the model, it must first be processed.

2.1.1 Getting Training and Testing Data

Since this study focuses on breast cancer, it utilizes normal breast samples (brca-rsem-fpkm-tcga and breast-rsem-fpkm-gtex) alongside tumor breast samples (brca-rsem-fpkm-tcga-t). The dataset consists of 1,071 tissue samples, each containing 19,738 gene features. Each gene is assigned a gene expression value, which measures the activity level of genes and proteins. Each gene has an expression for all tissues and serve as the model’s input data. The data is split into two sets: training (749) and testing (322), i.e., a 70/30 split. The data is split to ensure that both normal and tumor samples are represented in each set.

2.1.2 Data Preprocessing

The data processing replaces NA values, applies a $\log_2(n+1)$ transformation, and normalizes the data. Since gene expression values can be skewed, the log2 transformation reduces extreme values and stabilizes variance across the dataset. This transformation is commonly used in RNA-seq analysis to improve downstream results [10]. This research project and GemDIFF receives the training and test data in the same format [19].

2.1.3 SMOTE

Finally, due to the imbalance between normal and tumor samples, the Synthetic Minority Over-sampling Technique (SMOTE) is applied. SMOTE generates synthetic samples of the minority class (in this case, the normal samples) to balance the dataset. SMOTE is only applied to the training set after the split because the MLP needs sufficient representation of both classes to avoid bias toward the majority class, the tumor, when classifying [3].

2.2 Multilayer Perceptron (MLP) Classification Model

A Multilayer Perceptron (MLP) is a neural network following a feed-forward architecture. The neural network is an organization of layers where the data flows from the input layer to the hidden layers to the output layer. Each neuron (data point) in one layer connects to every neuron in the next, with an adjustable weight. Feed-forward occurs when data flows in one direction through each layer without looping back [12].

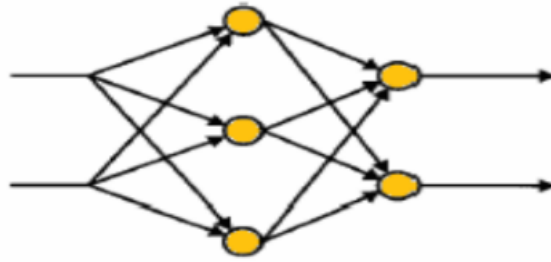


Figure 2.1: Example of MLP network "feed-forward" [12]

2.2.1 Forward Pass

During the forward pass, the input data is propagated through each layer of the neural network to create a prediction. As data moves through the network, it is multiplied by weights, and a bias is added. The weights are initialized to small random values, and the bias allows each neuron to shift its output. The weight sum is computed as [7]

$$z = Wx + b \quad (2.1)$$

Each weighted sum passes through a Rectified Linear Unit (ReLU), which introduces non-linearity to the neural network. If the weight sum is positive, it passes through; if it is negative, it is set to zero. ReLU can suppress inactive neurons and focus on active neurons, producing a positive signal [11].

2.2.2 Backpropagation

The key component of this research, and the reason why an MLP is used, is for backpropagation. After the input passes through the network and produces a prediction, the loss between the prediction and the actual value is computed. This error is then propagated backward through the neural network to identify how each weight contributes to the error. The model can adjust the weights and biases to minimize the loss function. Through this process, a gradient is created to each feature, allowing the model to find which genes influence the prediction [7].

2.2.3 Research Specific Model

For this research, the MLP model has an input layer with the number of features in the training set, 2 hidden layers with 256 and 64 neurons, and a last output layer with 2 neurons. The 19,738 gene expression values are fed into the neural network and gradually reduced through the hidden layer / ReLU until the output layer predicts whether the sample is a tumor or normal. Once the model is created, the training samples are used to evaluate the model's accuracy.

2.3 Gradient-Based Feature Attribution Model

The gradient-based feature attribution model extends the MLP by finding which particular genes influence the model’s prediction. Once the model is trained, it takes in the training data to predict whether a tissue sample is tumor or normal.

2.3.1 Sensitivity Analysis

To find what genes has the most influence, the method performs a sensitivity analysis by measuring how much the prediction (y) changes with the respect to adjusting the each gene’s expression (x):

$$R_i(x) = \frac{\partial y_c}{\partial x_i}(x) \quad (2.2)$$

2.3.2 Gradient x Input

High gradients shows that the gene expression has a strong influence while smaller gradients suggests little effect. A huge problem with the analysis is that it only measures change, and not the actual contribution. For example. if an expression is very small, but very high gradient, it does not significantly contribute to the model’s prediction. To counteract this, a Gradient x Input is computed instead by multiplying the gradient by the gene expression value:

$$R_i^c(x) = \frac{\partial S_c(x)}{\partial x_i} \cdot x_i \quad (2.3)$$

By performing this operation, we can measure how the model’s prediction changes with respect to the full contribution of each gene. The Gradient x Input method computes an attribution score vector for each gene feature in the tissue sample [1]. For example, in an arbitrary vector such as $[0.9, 0.1]$, the first feature has a higher influence on the prediction compared to the second feature. This 1D vector is created for all our training samples, forming a 2D Matrix.

2.3.3 Applying the Gradient Based Attribution Model

For this certain project, our goal is to find the genes that are correlated to breast cancer (tumors). So as the training computes their gradients, they get separated into two different buckets, tumor and normal. Each bucket creates a matrix of their corresponding attribution vectors, where

rows represents the sample and the column represents the genes.

To retrieve gene importance across samples, the mean attribution score is computed for both tumor and normal buckets. Since the distribution of attribution scores are not skewed, the mean is a reliable statistic for finding the gene importance.

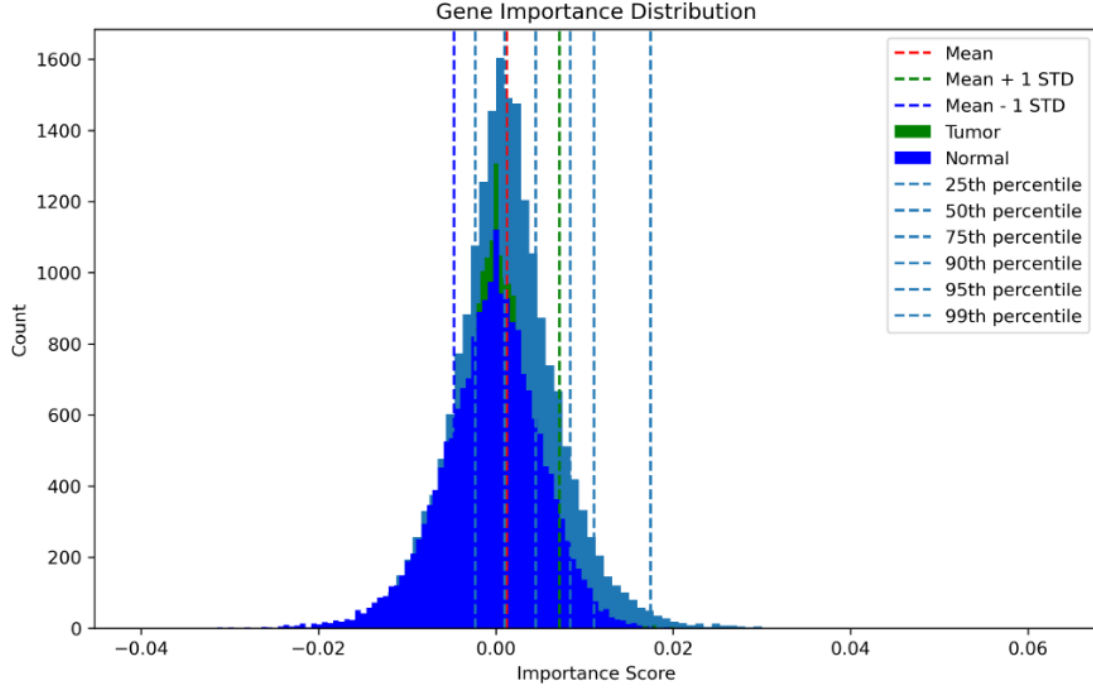


Figure 2.2: Distribution of Gene Importance Scores

$$\mu_{\text{tumor}} = \frac{1}{N_t} \sum R(x) \quad (2.4)$$

$$\mu_{\text{normal}} = \frac{1}{N_n} \sum R(x) \quad (2.5)$$

The importance score for each gene is the obtained by:

$$I = \mu_{\text{tumor}} - \mu_{\text{normal}} \quad (2.6)$$

The goal of this research is to identify genes associated with tumor samples. The mode rewards genes that contribute to breast cancer and reduce the genes that assist normal samples.

If a sample shows a strong attribution score for tumor and normal, the model can assume

that the gene is associated to biological processes, not particularly breast cancer. Similarly, if genes favor normal genes predictions, that feature does not correlate to breast cancer. This analysis prioritizes genes that contribute more to tumor predictions than normal predictions.

2.4 Feature Selection

While putting all gene features into an MLP classification model is the easiest approach, gene expression and phenotypic outcomes are complexly related. Datasets such as TCGA and GTEx are high-dimensional, and identifying biomarker genes within an interconnected web of layers makes it hard for a simple classification model to identify candidate genes[16].

To achieve the best results, the model has two different ways of extracting gene features for the MLP classification model.

- Input all 19,738 gene features
- Extract the silhouette score based off an MSigDB gene set (C4, C6, and Hallmark)

A silhouette score measures how well-separated clusters are; higher values indicate that samples have high cohesion and good separation from other clusters.

2.4.1 Silhouette Scores

GemDIFF uses a similar technique to determine whether tumor and normal samples separate in a UMAP-reduced space. They use silhouette scores to validate separation and confirm that there is something worth training [19]. On the other hand, this project uses silhouette scores to evaluate gene subsets (chunks) from MSigDB. By grouping genes that better separate, the MLP classification model can better detect signals and not treat all 19,738 features as noise.

The code finds the silhouette score using the built-in Scikit-Learn function. Given a set of genes, the function finds how well samples separate based on their gene expression values. Using the training data, cluster A contains other objects apart from a specific sample (i), then it computes:

$$a(i) = \text{average dissimilarity of } i \text{ to all other objects of A.} \quad (2.7)$$

When we have cluster B, which is the closet neighbor to A, it also computes:

$$b(i) = \text{average dissimilarity of sample } i \text{ to all samples in the nearest different class} \quad (2.8)$$

Then, the silhouette score is computed by:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (2.9)$$

Like before, the goal is to evaluate how well samples cluster based on selected genes [13].

2.4.2 Feature Extraction from MSigDB Gene Sets

MSigDB, the Molecular Signatures Database, is a widely used, comprehensive resource of gene sets for enrichment analysis [9]. These gene sets have coordinated biological activity across varying processes. In this work, there are three MSigDB collections used:

- C4 (computational gene sets, 1006 gene sets)
- C6 (oncogenic gene sets, 189 gene sets)
- H (hallmark gene sets, 50 gene sets)

The model computes silhouette scores for each gene set separately to extract meaningful gene features. First, the entire gene set is read, then divided into smaller subsets (chunks) of 16. If the number of genes is not divisible by 16, the run-over rolls into the final chunk. Afterward, the silhouette score for each is computed using the gene expression values from the training samples.

Once the scores are computed, they are then ranked, and those with scores above the mean plus one standard deviation are selected. All selected chunks better separate the tumor and normal samples. The corresponding genes in each selected chunk are then extracted and combined into a feature vector, which serves as the input to the MLP classification model. Rather than trying to classify all 19,738 genes, this method reduces the feature set so the model can focus on identifying meaningful signals.

In the end, the study compares gene lists generated from all features in the MLP with those extracted from silhouette scores in the C4, C6, and Hallmark gene sets.

2.5 Evaluation Methods

The last key method for this research project is analyzing the actual data for the breast cancer signal. Even though the model can find genes, further evaluation is needed to demonstrate that the methods and outputs are meaningful. There multiple different ways to see if whether our results are values including (1) Venn Diagrams, (2) Fisher’s Exact Test, (3) Gene Enrichment, and (4) Protein to Protein Interactions.

2.5.1 Venn Diagrams

Venn Diagrams are the most basic of our current evaluations, as they aim to provide a visual aid for comparing gene lists and showing any overlap. These diagrams allow quick comparison of shared and unique genes between sets. In this study, these diagrams will be used only to compare 2-3 sets. It is important to note that Venn diagrams do not provide any measure of statistical significance. They serve only as an exploratory tool to provide a quick overview between sets.

2.5.2 Fisher’s Exact Test

The Fisher’s Exact Test is the first method in this study that provides a form of measurement for statistical significance. The test evaluates the association between two categorical variables in a 2×2 contingency table [5]. The null hypothesis states that there is no association between the two variables in the contingency table. When $p < 1 \times 10^{-10}$, it is considered highly statistically significant and rejects the null hypothesis, whereas a high p-value indicates a failure to reject the null hypothesis. Typically, when a p-value is less than 0.05 it is significant, but the results should be extremely accurate when working in the medical field.

Fisher’s exact test is used to determine if the overlap between two gene sets is statistically significant, rather than occurring by chance. The test is applied to a 2×2 contingency table constructed from the number of genes shared by the sets, unique to each set, and present in neither set. Only the p-value was used to evaluate statistical significance.

	In <i>set2</i>	Not in <i>set2</i>
In <i>set1</i>	<i>a</i>	<i>b</i>
Not in <i>set1</i>	<i>c</i>	<i>d</i>

Table 2.1: Contingency table used for Fisher’s exact test.

In this analysis, there are a couple of different gene sets:

- All Features in MLP
- C4
- C6
- Hallmark
- Core307
- Random307
- Universal

Core307 is a gene set identified in prior work and shown to enrich for breast cancer. It is created by preprocessing the data, as in this study, followed by clustering genes into groups on a UMAP plot. Each cluster is then scored using silhouette scores to identify separated gene sets, which are used to determine the top 307 genes that perform best in a diffusion model [19]. Random307 is a random set of 307 genes within the same study. The universal gene set is every single gene (19,738).

To start the analysis, the test takes in two different gene sets. All of the gene sets identified by the MLP is compared against both Core307 and Random307. For each test, a 2 x 2 contingency table is created, consisting of genes shared between the sets, the genes unique to each list, and genes absent from both sets in the Universal set.

The Fisher’s exact test computes two values: an odds ratio and p-value. The odds ratio measures the association between two variables by comparing the probability of an event occurring in one group with respect to another. For this study, the odds ratio is not the primary focus. Instead, we look more at the p-value, which represents the probability of observing the overlap or extreme set under the null hypothesis.

$$P = \frac{\binom{a+b}{a} \binom{c+d}{c}}{\binom{n}{a+c}} \quad (2.10)$$

Depending on the gene sets being compared, the statistical test should produce different results. If the model is comparing a set with Random307, the p-value is expected to be high, representing no significant overlap. When the p-value is low, it implies greater overlap than expected with a set not associated with breast cancer. This may suggest that the MLP is not extracting a meaningful cancer signal, but more non-specific genes.

In contrast, when the model is comparing a set against Core307, an established breast cancer gene set, the p-value should be very low to indicate significant overlap. This suggests that the gene set identifies genes associated with breast cancer. Having a low p-value against Core307 does not imply that the gene set is fully enriched for breast cancer; only that there is significant overlap.

2.5.3 Gene Enrichment

Gene Enrichment is the primary analysis for each gene set. The analysis is performed using ToppFun from the ToppGene Suite, which detects functional enrichment in gene lists to find Transcriptome, Proteome, Regulome (TFBS and miRNA), Ontologies (GO, Pathway), Phenotype (human disease and mouse phenotype), Pharmacome (Drug-Gene associations), literature co-citation, and other features. ToppFun identifies "significantly enriched functional annotations" by performing statistical tests to detect meaningful overlap among gene sets and computing p-values. A low p-value indicates that the overlap is not due to chance [4].

The enrichment results are presented as a large table, in which the key categories to look at are:

- Disease (e.g., breast cancer-related processes)
- Drugs (e.g., drugs targeting breast cancer)
- Pathways (groups of genes that carry out specific biological processes)

Since the analysis yields many enriched terms, a cutoff is needed to identify only significant terms. For this study, the "q-value FDR B&H" (Benjamini-Hochberg method) value must be $< 1 \times 10^{-10}$.

The q-value, or false discovery rate, is the p-value adjusted for multiple tests. It estimates the expected proportion of false positives, so to obtain meaningful results, these values must be low to indicate that the results are statistically reliable.

2.5.4 Protein to Protein Interaction

The last evaluation method is to conduct a protein–protein interaction (PPI) analysis. The goal of this analysis is to create a topology graph using the STRING database to construct a network of protein–protein associations derived from experimental assays, computational predictions, and prior knowledge[15].

Using the STRING API, the code takes the defined gene sets and maps them to their corresponding proteins in *homo sapiens*. It then constructs a network in which each protein is represented as a node and each interaction as an edge. Once the nodes and edges are identified, the analysis measures connectivity, which is the average number of interactions per protein:

$$\text{connectivity} = \frac{2 \times \# \text{ of edges}}{\# \text{ of nodes}} \quad (2.11)$$

For the connectivity to be significant, the same analysis is repeated 100 times using randomly generated gene sets of equal size to the original set. If the original connectivity is higher than then average connectivity of the random gene sets, this indicates that the original gene set is more connected than if it were to happen by chance. With a higher connectivity, the more related the proteins are. The p-value can then be extracted using the empirical p-test:

$$p = \frac{\#(\text{random} \geq \text{observed})}{\text{total random samples}} \quad (2.12)$$

For the results to be significant, the p-value should be less than 0.05 [8].

Chapter 3

Related Work

3.1 Gene signatures provides prognostic information in breast cancer

Previous work has shown that gene expression signatures play a major role in understanding and predicting breast cancer outcomes. Established gene signatures, such as the 21-gene and 70-gene panels, provide significant prognostic information and guide treatment decisions in breast cancer patients. These gene sets are derived from large-scale datasets and capture meaningful biological patterns related to tumor development [2].

This relates directly to this thesis, as both approaches aim to identify biologically relevant gene sets in breast cancer. While prior work focuses on predefined gene signatures found through clinical and statistical methods, this thesis leverages artificial intelligence and computational approaches to extract gene features from high-dimensional data. By allowing a model to analyze the data directly, the process becomes faster and more scalable compared to traditional methods. Since gene expression has already been shown to be a valid approach for identifying breast cancer-related genes, applying this method may help support future treatment decisions.

3.2 Biomarker Genes with enriched classification from Hallmark gene sets

A previous study has already used Hallmark gene sets and classification to identify biomarker genes within these sets. Utilizing the same TCGA and GTEx data, neural networks are applied to high-dimensional gene expression data to extract key genes. The main difference is that, rather than creating gradient-based attribution scores, it evaluates how well the gene sets classify samples. Gene sets with higher classification performance are considered more biologically meaningful and are then used to identify important genes within those sets [16].

This research provides a more focused approach by directly evaluating the classification model’s predictions rather than the entire gene set’s performance. Instead of simply determining which gene set performs best, this study fine-tunes the analysis by ranking individual genes by importance.

3.3 Gene Selection and Classification of Cancer using Deep Learning

Previous work has combined deep learning with feature selection to identify important genes from high-dimensional gene expression data. In particular, the study highlights that these large gene expression datasets often exhibit high complexity, underscoring the importance of feature selection. Their approach uses neural networks and feature selection methods, similar to this study, to classify data and reduce noise in the dataset [14].

Both studies aim to extract meaningful sets of genes to determine which genes best classify cancer. However, their work focuses on improving classification accuracy, rather than validating the biological relevance of the selected genes. Their feature selection approach uses all gene features and applies metaheuristic optimization methods to select a subset of genes that reduces noise. In contrast, this study uses silhouette scores to identify gene subsets and further evaluates these sets using connectivity and biological validation, providing a more comprehensive assessment of gene relevance.

Chapter 4

Results

This section highlights the results found through the process highlighted in the Research Design and Methods chapter. As the paper goes through varying methods and gene sets, some gene sets may be loss due to insignificance. The goal is to extract the set of gene that best associate with breast cancer.

As a reminder, there are a couple of different methods in for choosing which features to input for the MLP. All 19,738 gene features along side the Silhouette Score (greater than threshold of mean + 1 standard deviation) of C4, C6, and Hallmark. Entering the silhouette score analysis, there are 10,749 unique genes for C4, 10,927 for C6, and 4384 for Hallmark.

The results are also associated with Core307, a gene set in a previous research that correlates to breast cancer using a different methodology, and Random307, 307 random genes.

4.1 MLP Classification Model Accuracy

As each gene set enters the MLP classification model, it outputs an accuracy:

Class	Precision	Recall	F1-score	Support
Normal	0.94	1.00	0.97	31
Tumor	1.00	1.00	1.00	291
Accuracy	0.994			322

Table 4.1: Test set performance metrics for the MLP classifier using 19,738 features

Class	Precision	Recall	F1-score	Support
Normal	0.97	1.00	0.98	31
Tumor	1.00	1.00	1.00	291
Accuracy	0.997			322

Table 4.2: Test set performance metrics for the MLP classifier using 4,311 features extracted from the C4 gene set.

Class	Precision	Recall	F1-score	Support
Normal	0.91	1.00	0.95	31
Tumor	1.00	0.99	0.99	291
Accuracy	0.991			322

Table 4.3: Test set performance metrics for the MLP classifier using 3,311 features extracted from the C6 gene set.

Class	Precision	Recall	F1-score	Support
Normal	0.91	1.00	0.95	31
Tumor	1.00	0.99	0.99	291
Accuracy	0.991			322

Table 4.4: Test set performance metrics for the MLP classifier using 968 features extracted from the Hallmark gene set.

Despite the MLP classifier achieving high accuracy values, they should be interpreted with caution, as the high value could be due to class imbalance. It is important to consider other measurements, such as the F1-Score, that account for imbalance, as they balance precision and recall.

Epoch vs Loss C4

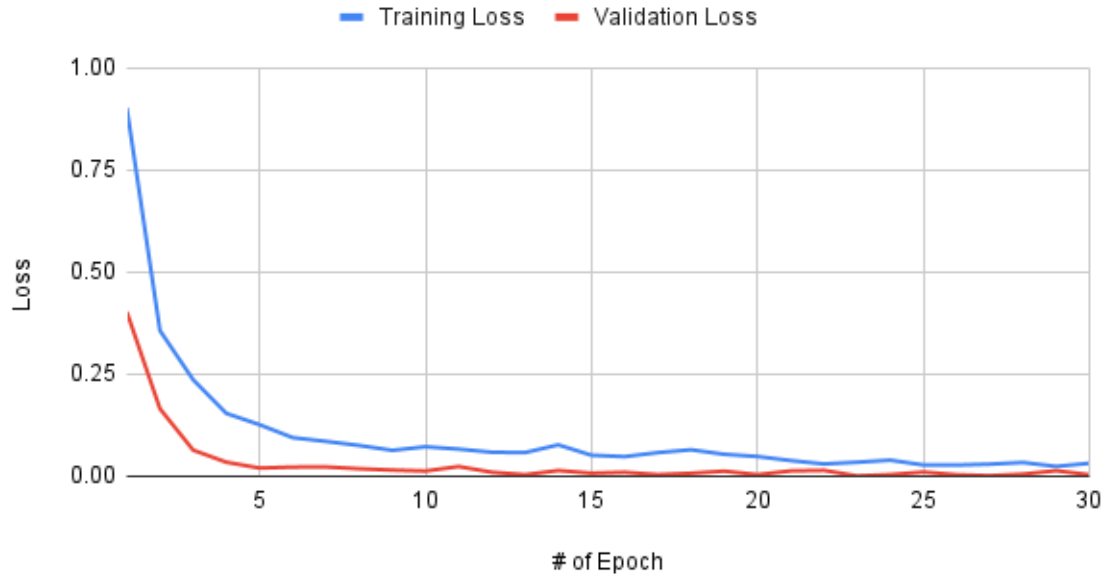


Figure 4.1: Training and Validation Loss vs. Epoch (C4 gene set)

The figure above compares the training and validation losses at each epoch for the C4 gene set. Typically, high accuracy is a huge concern with overfitting, which happens when the model learns the data rather than how to predict. The figure above shows the training and validation losses over each epoch, which are metrics that evaluate how a model performs during the learning process. Since the training and validation losses decrease together at the same rate with a minimal gap, there appears to be no overfitting [18].

4.2 Utilizing all 19,738 Gene Features - MLP307

4.2.1 Venn Diagram

Once the models extracts a list of genes it finds associated with breast cancer, an analysis is made to compare the genes between two other sets, Core307 and Random307. When comparing between the two sets, the analysis wants to find a significant overlap between Core307 and no significance between Random307. This analysis will provided insight on whether or not there is a pattern on the results to further continue the study.

Venn Diagram: MLP307 vs Core307 vs Random307

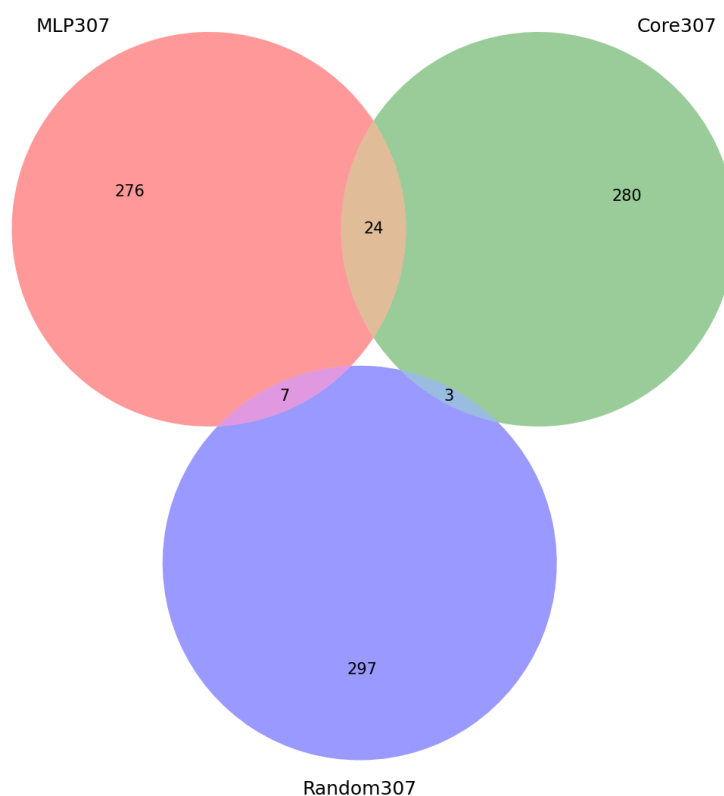


Figure 4.2: Venn Diagram comparing the gene sets of MLP307, Core307, and Random307

The first analysis is to create a three-way Venn diagram comparing MLP307, Core307, and Random307. Looking at the analysis itself, there is a clear indication that MLP307 vs. Core307 (24) has a larger overlap than MLP307 vs. Core307 (7). This hints that there is some significance between MLP307 and Core307.

To further prove this point, when doing 100 iterations and generating a 307-random-gene list, the average overlap between MLP307 vs Random307 is 4.77, and between Core307 vs Random307 is 4.73. This small difference in average overlap shows that the overlap between MLP307 and Core307 is higher than what would be expected if it were random, but there still needs to be further testing to actually get a quantitative value.

4.2.2 Fisher's Exact Test

Once the Venn Diagram is created, the analysis needs quantitative p-values to show for significant comparisons.

Comparison	Odds Ratio	P-value
MLP307 vs Core307	5.737	9.92×10^{-11}
MLP307 vs Random307307	1.488	2.03×10^{-1}

Table 4.5: Fisher's exact test results comparing overlap between MLP307, Random307, and Core307.

In the table above, a Fisher's exact test is performed to compare MLP307 and Core307. Since this study is conducted in a medical context, a significant overlap is defined when $p < 1 \times 10^{-10}$. In this case, the overlap between MLP307 and Core307 is below this threshold, indicating statistical significance. This low p-value does not imply that the identified gene list is directly associated with breast cancer, but rather that there is a relationship with an already established gene set. Additionally, the p-value between MLP307 and Random307 is much higher, suggesting that there is insufficient evidence to conclude that MLP307 is a randomly generated gene list.

Again, to prove that the Fisher's exact test is extracting real data, an analysis is made to compare five more random gene sets against Core307 and Random307.

Comparison	Odds Ratio	P-value
Random307_1 vs Core307	1.049	5.21×10^{-1}
Random307_1 vs Random307	1.049	5.21×10^{-1}
Random307_2 vs Core307	0.411	9.54×10^{-1}
Random307_2 vs Random307	0.411	9.54×10^{-1}
Random307_3 vs Core307	2.169	2.25×10^{-2}
Random307_3 vs Random307	0.204	9.92×10^{-1}
Random307_4 vs Core307	0.833	7.06×10^{-1}
Random307_4 vs Random307	1.267	3.44×10^{-1}
Random307_5 vs Core307	1.488	2.03×10^{-1}
Random307_5 vs Random307	0.411	9.54×10^{-1}

Table 4.6: Fisher's exact test results comparing random gene sets (Random307) with Core307 and other random sets.

When examining the p-values for each random gene set, they are much greater than 1×10^{-10} , indicating that the random gene sets are not showing a correlation with Core307. This supports that MLP307 is significant and not randomly generated.

4.2.3 Threshold Analysis

Another analysis, unique to the MLP307 section, is performed by not only extracting the top 307 genes, but also comparing gene sets whose importance scores are above a certain threshold (25, 50, 75, 90, 95, 99).

The goal is to show that the genes selected by the MLP are not limited to a fixed subset, but instead consistently demonstrate meaningful relationships across different score thresholds.

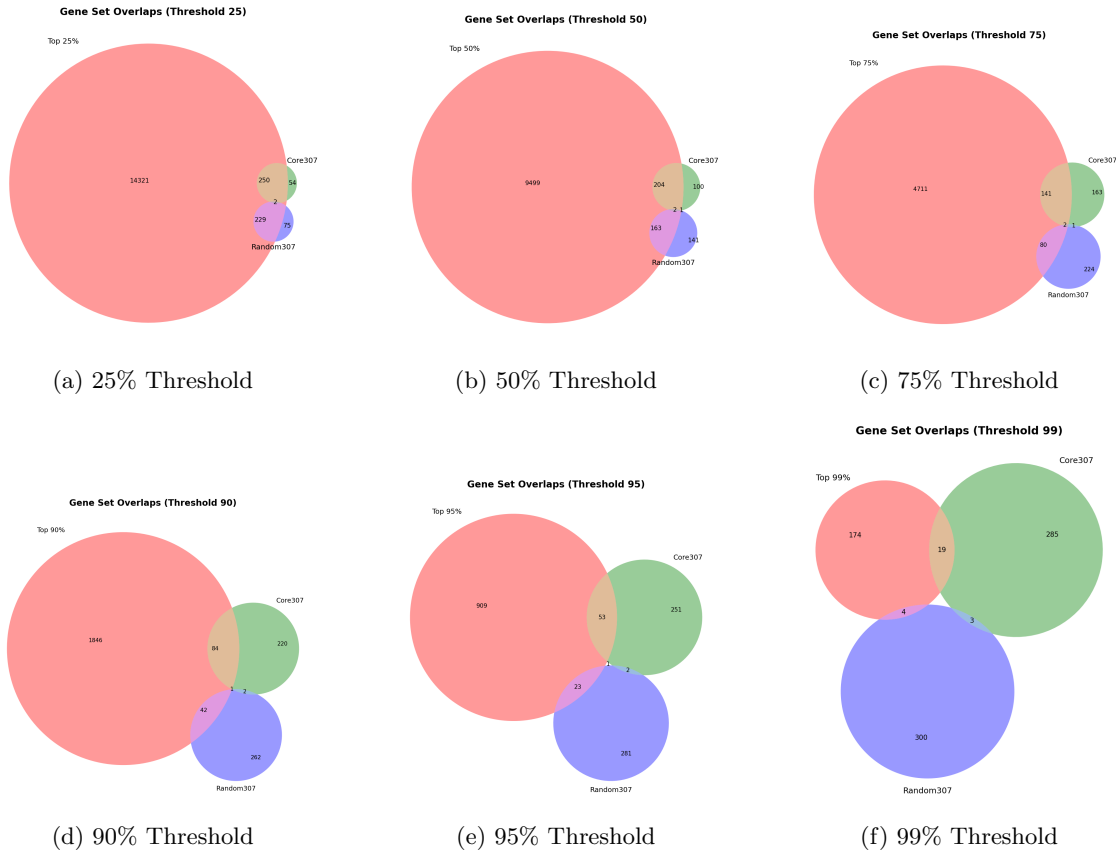


Figure 4.3: Comparison of Gene Sets Selected from 19,738 Genes Above a Threshold Against Core307 and Random307

The table above shows the progression of altering the MLP importance threshold when comparing gene sets against Core307 and Random307. As the p-value suggests, as the threshold in-

Threshold	Comparison	Odds Ratio	P-value
99	Threshold 99 vs Core307	7.135	2.77×10^{-10}
99	Threshold 99 vs Random307	1.316	3.67×10^{-1}
95	Threshold 95 vs Core307	4.236	4.49×10^{-16}
95	Threshold 95 vs Random307	1.628	2.05×10^{-2}
90	Threshold 90 vs Core307	3.557	9.56×10^{-19}
90	Threshold 90 vs Random307	1.477	1.47×10^{-2}
75	Threshold 75 vs Core307	2.664	1.32×10^{-16}
75	Threshold 75 vs Random307	1.095	2.62×10^{-1}
50	Threshold 50 vs Core307	2.062	7.73×10^{-10}
50	Threshold 50 vs Random307	1.165	1.03×10^{-1}
25	Threshold 25 vs Core307	1.536	1.80×10^{-3}
25	Threshold 25 vs Random307	1.013	4.91×10^{-1}

Table 4.7: Fisher’s exact test results comparing threshold-based gene sets with Core307 and Random307.

creases, the correlation between Core307 becomes stronger, while that between Random307 remains relatively unchanged.

At lower thresholds (e.g., 25), the gene set is extremely large (14,803 genes), which includes many low-ranking importance scores. This broad amount of genes results in weaker statistical significance. As the threshold increases, prioritizing the more important genes yields a stronger correlation. In contrast, the comparison between Random307 indicates no change, regardless of gene set size. This indicates that the MLP Classification Model is finding a signal, not finding random genes.

4.2.4 Gene Enrichment

Lastly, for the MLP307 gene set, it is necessary to determine whether it is specifically enriched for breast cancer-related functions. Although the gene set shows significant overlap with a pre-established gene set, the biological pathways and interactions may correspond to a different function. To test this, a ToppFun functional enrichment analysis is used to identify the biological processes represented.

When analyzing and filtering for a q-value $\text{FDR B\&H} < 1 \times 10^{-10}$, only 37 pathways were identified, a majority of which came from the ToppCell Atlas category. There was no enrichment for any breast cancer disease or drug pathways. While the ToppCell Atlas category includes information on breast cancer, these results are too broad, focusing on cell types rather than specific disease processes. For the analysis to enrich for breast cancer, there needs to be enriched terms on the

diseases and drug pathways. In the end, the MLP307 gene set does not enrich for breast cancer-related biology.

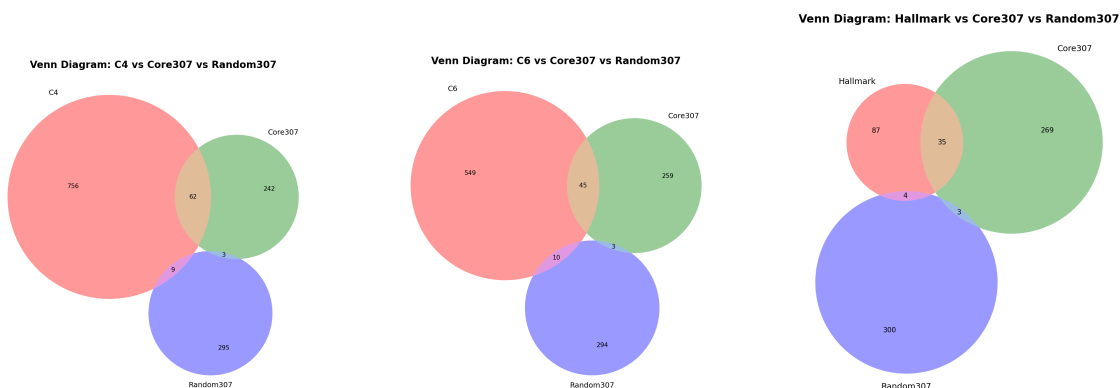
4.3 Utilizing Gene Features from C4, C6, and Hallmark Gene Sets

Although using all gene features in the MLP classification model did not show a significant result, the next part attempts to identify significant genes using fewer features, as discussed in Chapter 2.4. Including all 19,738 gene features may introduce significant noise or make the analysis overly complex. So, filtering the gene set with key features could fix this issue.

After independently extracting the genes from C4, C6, and Hallmark whose cutoff scores are one standard deviation above the mean, and then extracting the genes whose importance score is one standard deviation above the mean, a gene set is obtained. Using a threshold approach rather than selecting a fixed number provides a more statistically meaningful cutoff.

4.3.1 Venn Diagram + Fisher's Exact Test

Similar to MLP307, the goal of this section is to determine whether there is a significant overlap between each gene set and Core307 and little to no overlap between Random307. The Venn Diagram provides a visual representation of the overlap, whereas Fisher's exact test provides the statistical validation.



(a) Venn Diagram comparing the gene sets of C4 (827 gene features), Core307, and Random307

(b) Venn Diagram comparing the gene sets of C6 (604 gene features), Core307, and Random307

(c) Venn Diagram comparing the gene sets of Hallmark (126 gene features), Core307, and Random307

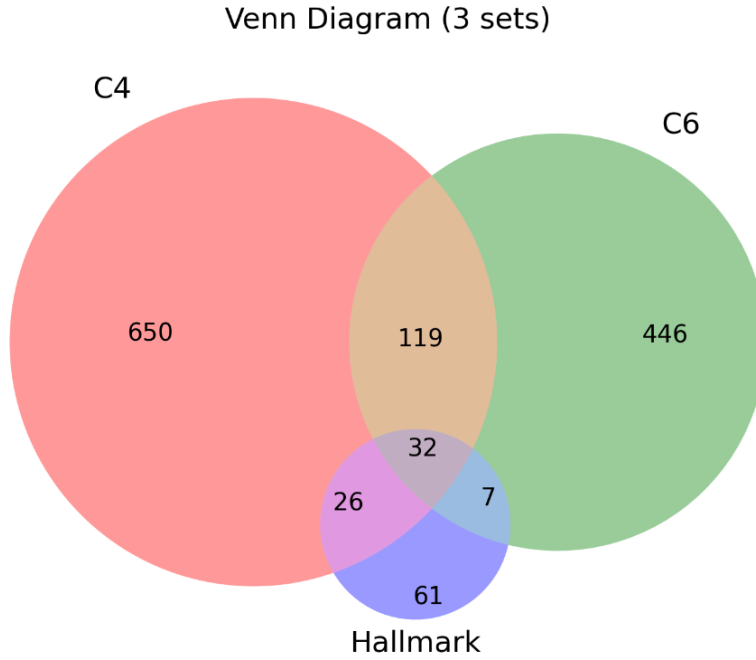


Figure 4.5: Venn Diagram comparing the gene sets of C4, C6, and Hallmark

Comparison	Odds Ratio	P-value
C4 vs Core307	6.174	1.82×10^{-25}
C4 vs Random307	0.687	0.69×10^{-1}
C6 vs Core307	5.798	2.01×10^{-18}
C6 vs Random307	1.068	4.66×10^{-1}
Hallmark vs Core307	27.344	3.54×10^{-34}
Hallmark vs Random307	2.089	2.09×10^{-1}

Table 4.8: Fisher's exact test results comparing overlap between gene sets C4, C6, and Hallmark against Random307 and Core307.

When looking at the p-value for each comparison, all gene sets show a significant overlap between Core307 ($p\text{-value} < 1 \times 10^{-10}$), while it fails to show a significant overlap between Random307. This indicates that the found gene sets do correlate with a biologically established gene set.

4.3.2 Gene Enrichment

Disease Term	P-value
Liver carcinoma	3.26×10^{-47}
Malignant neoplasm of breast	1.44×10^{-20}
Mammary Neoplasms	5.50×10^{-20}
Breast Carcinoma	1.51×10^{-19}
Mammary Carcinoma, Human	2.18×10^{-19}
Mammary Neoplasms, Human	2.18×10^{-19}
Prostatic Neoplasms	1.30×10^{-18}

Table 4.9: Top enriched disease terms in C4 gene set.

Disease Term	P-value
Liver carcinoma	4.14×10^{-33}
Malignant neoplasm of prostate	9.62×10^{-14}
Prostatic Neoplasms	9.62×10^{-14}
Malignant neoplasm of breast	2.86×10^{-13}

Table 4.10: Top enriched disease terms in C6 gene set.

Disease Term	P-value
Liver carcinoma	2.15×10^{-44}
Malignant neoplasm of breast	1.91×10^{-24}
Mammary Carcinoma, Human	2.03×10^{-24}
Mammary Neoplasms, Human	2.03×10^{-24}
Mammary Neoplasms	2.39×10^{-24}
Breast Carcinoma	5.73×10^{-24}
Urinary bladder cancer (marker)	9.61×10^{-15}

Table 4.11: Top enriched disease terms in Hallmark gene set.

The enrichment analysis identifies breast cancer-related diseases within each gene set and highlights drugs used for breast cancer, such as palbociclib and doxorubicin. Looking further into

the enrichment, there appear to be fewer breast cancer-related drugs in Hallmark compared to C4 and C6, notably fulvestrant or paclitaxel. Although Hallmark does enrich for breast cancer disease terms, the remainder of this study will focus on C4 and C6, as their original gene sets are made to represent cancer rather than general biological processes.

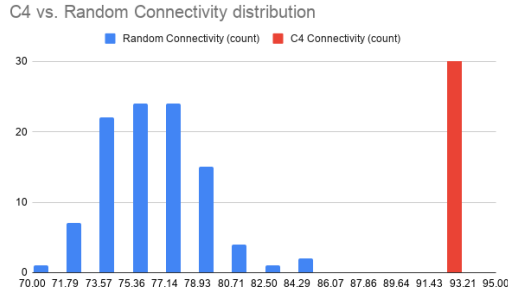
4.3.3 C4 and C6 Gene and Protein Interaction

As the study continues, the initial thought was to take the intersection between C4 and C6 (151 genes) and perform an enrichment analysis. The idea is that because C4 and C6 both correlate to breast cancer, the overlap should as well. However, the enrichment only showed Liver carcinoma as a disease. Although the enrichment included drugs correlated with breast cancer, the lack of disease terms suggested that one of the gene sets may not have a strong correlation as originally thought.

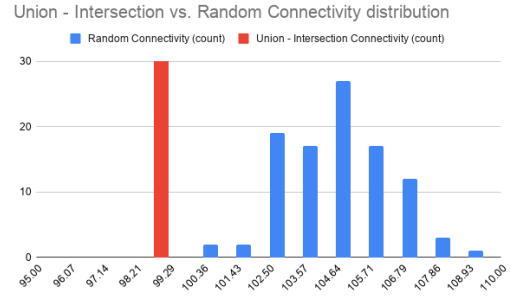
Further analysis showed that the Union - Intersection (all unique genes in C4 and C6) contained the missing breast cancer terms. To better understand this issue, a protein-protein interaction analysis was performed on C4, C6, the intersection, and Union - Intersection. This analysis creates a topology graph and evaluates the connectivity between the genes. The same analysis is then performed on random gene sets of equal size, repeated 100 times. The goal is to determine whether the observed gene sets have higher connectivity than random sets, indicating statistically significant results.

	# of Genes	# of Edges (PPI)	Connectivity (2 * edges / nodes)
C4	827	38314	92.66
C6	604	15180	50.26
C4 and C6 Intersection	151	2050	27.15
C4 and C6 Union - Intersection	1129	55689	98.65

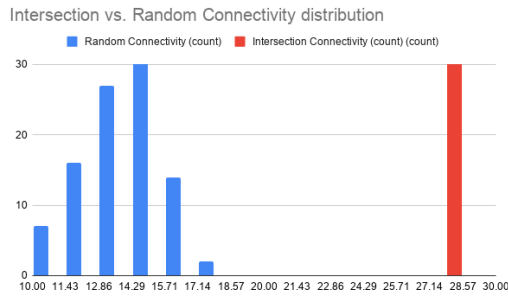
Table 4.12: Protein-Protein Interaction between C4 and C6



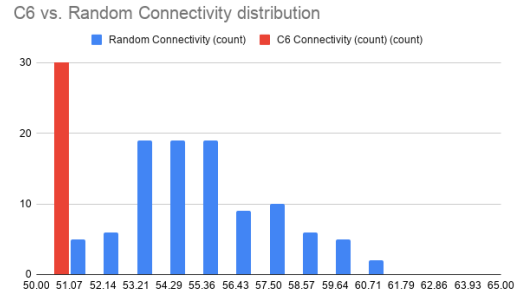
(a) C4 vs. Random Connectivity distribution



(b) Union - Intersection vs. Random Connectivity distribution



(c) Intersection vs. Random Connectivity distribution



(d) C6 vs. Random Connectivity distribution

Figure 4.6: Connectivity distributions compared to random gene sets

As the results show, both C4 and the intersection of C4 and C6 show higher protein-protein interaction connectivity compared to the random distribution, as their connectivity values are above the distribution. In contrast, C6 and the union-intersection are both below the distribution.

Since C4 shows this pattern while C6 does not, it suggests that C6 may have weaker biological interactions within the set. The fact that the intersection is still above the distribution shows that the overlapping genes retain strong protein-protein interaction, but this does not appear to be specific to breast cancer and may instead reflect more general biological processes.

In the end, C4 is the only independent gene set that shows a clear association with breast cancer.

Chapter 5

Conclusions and Discussion

This thesis presents an analysis of identifying a gene set enriched for breast cancer, outlining the process from data collection to feature extraction and implementation in an MLP classification model. This chapter summarizes the final interpretation of the findings, outlines limitations, and identifies directions for future work.

5.1 Interpretation of Evaluation Findings

As stated at the start of the paper, the goal is to find a gene set enriched for breast cancer. As the study progresses, the gene sets are refined using a series of statistical analyses.

The first evaluation used all 19,738 genes as features in the MLP classification model. The top 307 genes extracted showed statistical overlap with a pre-established gene set and no statistical overlap with a random gene set. Despite this finding, the gene set did not enrich for breast cancer pathways and could not continue for further analysis. However, the analysis still showed meaningful statistics on the MLP's performance. The classification model is capturing real biological signals rather than selecting genes at random, so additional sets or selected features may fix the issue.

Next, the MLP was applied again using a more refined gene-feature set. Using MSigDB gene sets and filtering for those with a silhouette score greater than one standard deviation above the mean, the goal was to reduce noise and model complexity. The main gene sets utilized were C4, C6, and Hallmark. Fisher's exact test showed that each set had statistically significant overlap against the pre-established breast cancer gene set.

However, further enrichment analysis showed that the Hallmark set was only weakly relevant to breast cancer. Because of its smaller size and weaker enrichment, the focus shifted to C4 and C6, which showed stronger signals. A concern appeared when the intersection of C4 and C6 enriched for liver disease instead of breast cancer.

To investigate this further, protein–protein interaction analysis was performed. The results showed that C6 had weaker connectivity than expected by chance, while C4 remained above the random distribution. The intersection between C4 and C6 was statistically higher than its random distribution counterpart, meaning that the enrichment still showed some level of connectivity and was not purely noise. Based on these results, while the C6 gene set is not recognized as being enriched for breast cancer, C4 is, as it consistently passed all stages of the analysis.

5.2 Limitations

The key limitations of this thesis include the accuracy of the MLP classification model, the scope of feature extraction, and the lack of additional validation analyses.

First, the model achieving a high accuracy should be interpreted with caution. While the thesis showed that overfitting did not appear to be an issue, the imbalance between normal and tumor samples may introduce bias or uncertainty in the model’s performance. A more balanced dataset could reduce this uncertainty. Additionally, while this study used 256, 64, and 2 output layers and demonstrated that it worked, the rapid reduction in dimensionality may cause the model to lose important features or rely on a few dominant signals.

Second, while C4 showed statistically meaningful results across each analysis in this study, the feature extraction process was limited. This study only focused on MSigDB gene sets, but there are many other open-sourced gene sets and alternative methods to silhouette scores that could be used to extract key features. Additionally, the fact that the overlap between C4 and C6 does not enrich for breast cancer highlights a potential issue, as both sets follow a similar extraction process.

Lastly, the study lacks additional validation methods to ensure that C4 truly correlates with breast cancer. The study shows that there are biological pathways related to breast cancer, yet there still could be differences when comparing to current studies or actual implementation.

5.3 Future Work

There are several ways to follow up on this thesis.

First, future work can include collecting a more balanced dataset between normal and tumor samples to reduce potential bias in the classification model.

Second, additional feature extraction methods can be explored beyond MSigDB gene sets, including other public gene databases and alternative clustering or scoring techniques. This could help identify more biologically relevant gene sets.

Lastly, further validation analyses can be performed, such as testing on other datasets or incorporating biological validation methods to confirm that the identified genes, particularly C4, correlate with breast cancer.

5.4 Concluding Remarks

This thesis demonstrates that a gene set correlating towards breast cancer is possible and is supported through a series of statistical analyses. By processing genomics data, filtering key features, and applying the filtered data into a classification model, the extracted 827 genes from the C4 gene set showed strong interaction and correlation with breast cancer.

Beyond identifying biological processes, this thesis highlights the use of artificial intelligence within the medical field by combining machine learning with traditional biological analysis methods. This approach not only improves the efficiency of identifying meaningful gene sets but also provides a framework for future studies in disease analysis and gene expression.

5.5 Acknowledgments

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